

Geometry

Unit 9

Lesson 9 Homework

Name____**Answer Key**_____

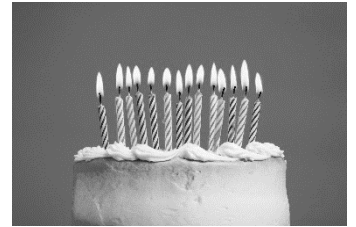
Date_____

Period_____

1. Brea enjoys baking and decorating cakes. The easiest cake pan creates a cake that is 9 inches in diameter and $4\frac{1}{4}$ inches tall. What is the surface area of the cake? Round to the nearest hundredth.

$$SA = 2\pi\left(4\frac{1}{2}\right)\left(4\frac{1}{4}\right) + 2\pi\left(4\frac{1}{2}\right)^2 = 38.25\pi + 40.5\pi$$

$$= 78.75\pi \approx 247.40 \text{ in}^2 \text{ (when using the pi button)}$$



2. Brea ices every possible space of the cake. She only adds her creative designs to the top of the cake. What is the area she has for her creative designs? Round to the nearest hundredth.

$$A = \pi\left(4\frac{1}{2}\right)^2 = 20.25\pi \approx 63.62 \text{ in}^2 \text{ (when using the pi button)}$$

3. A baseball has a diameter of 7.5 centimeters. What is the surface area of a baseball? Round to the nearest hundredth.

$$SA = 4\pi(3.75)^2 = 56.25\pi \approx 176.71 \text{ cm}^2 \text{ (when using the pi button)}$$



4. The diameter of a softball is 8.89 centimeters. What is the surface area of a softball? Round to the nearest hundredth.

$$SA = 4\pi(4.445)^2 = 79.032\pi \approx 248.29 \text{ cm}^2 \text{ (when using the pi button)}$$

5. What is the difference in surface area between a baseball and a softball?
 $248.29 - 176.71 = 71.58 \text{ cm}^2$

6. Newman's fence needs a new coat of paint. The fence runs along the east side of the property line which is 120 feet long. The height of the fence is 15 feet with a width of 0.5 feet. Both sides of the fence and the top need to be painted. Find the area of the fence that needs to be painted.

$$A = 2(120)(15) + (120)(0.5) = 3600 + 60 = 3660 \text{ ft}^2$$



7. If 1 gallon of paint covers 410 square feet, how much paint does Newman need to buy?

$$3660 \div 410 \approx 8.93$$

Newman needs to buy 9 gallons.

8. If 1 gallon of paint costs \$23.20, how much will the paint cost?

$$9 \times 23.20 = \$208.80$$

9. Gertrude wants to build a new pen for her cow. She has one area of her property available for a rectangular pen that would measure 18 feet by 36 feet. She has a second area of her property available for a square pen that would measure 25 feet wide. Which pen area would require the least amount of fence?

$$\text{Pen 1} = (18 + 36)(2) = 108 \text{ ft}$$

$$\text{Pen 2} = 25 \times 4 = 100 \text{ ft}$$

Pen 2 would require the least amount of fence.



10. Gertrude decides to build the pen area with the least amount of fence. If her fence is 7.5 feet tall, what is the surface area of her fence?

$$(7.5)(100) = 750 \text{ ft}^2$$